

## MODULE 11

# SQL Views and Virtual Tables

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DBMS with SQL - Enterprise Learning Series

ARSHITH LMS | Professional Edition

# SQL Views and Virtual Tables

## What is a View?

A saved query acting as a virtual table.

Does NOT store data itself.

Benefits: simplify queries, improve security, consistent interface.

## Create View

```
CREATE VIEW student_summary AS
SELECT first_name, department, gpa
FROM students WHERE status = 'active';

SELECT * FROM student_summary WHERE gpa > 3.5;
```

## Replace and Drop

```
CREATE OR REPLACE VIEW student_summary AS
SELECT first_name, gpa FROM students;

DROP VIEW IF EXISTS student_summary;
```

## Updatable Views

Single base table, no GROUP BY, no aggregates.

```
UPDATE student_summary
SET gpa = 3.9 WHERE last_name = 'Kumar';
-- Updates underlying students table
```

## Security with Views

```
CREATE VIEW public_emp AS
SELECT emp_id, name, department FROM employees;
-- salary column is hidden!

GRANT SELECT ON public_emp TO 'hr'@'localhost';
```